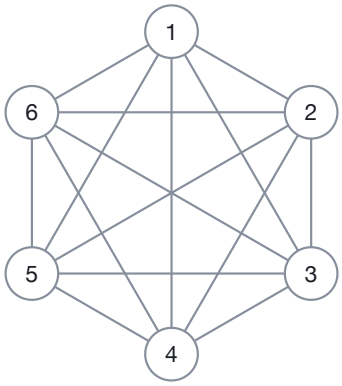
EXPLAINER

Pair-by-pair links grow much faster than the systems

Each new system needs a link to every system already there; through a hub it needs one.

Pair by pair

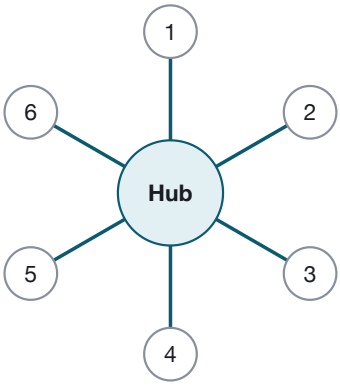


15 links

= 1 + 2 + 3 + 4 + 5

= n(n – 1) ÷ 2 = 6 × 5 ÷ 2

Through one hub



6 connections

= 1 + 1 + 1 + 1 + 1 + 1

= n = 6

1. One system on its own needs no links. Through a hub, it needs one connection.
2. A second system: pair by pair it needs one link, to the first. Through the hub it adds one connection.
3. A third system must link to both systems already there: 2 new links, 3 in all. The hub still adds 1.
4. A fourth system adds 3 links, one to each existing system: 6 in all. The hub adds 1: 4 in all.
5. A fifth system adds 4 links: 10 in all, against 5 connections.
6. A sixth adds 5 links: 1 + 2 + 3 + 4 + 5 = 15, against 6 connections.
7. The sum is n(n – 1) ÷ 2: each of the n systems links to the n – 1 others, and every link is counted from both ends, so halve it. Through the hub it is just n.

What this shows. With n systems, linking every pair needs 1 + 2 + ... + (n – 1) = n(n – 1) ÷ 2 links, each one built, tested and kept working. Through one hub each system connects once, so n connections. At 20 systems that is 190 against 20.

What this leaves out: each connection still needs an owner's decision about what it may carry; the arithmetic counts integration work, not approvals.

Numbers computed from n(n – 1) ÷ 2 and n.